

Demonstration of Proposed Features

Date	2 July 2026
Team ID	
Project Name	Smart Lender
Maximum Marks	1 Mark

Demonstration of Proposed Features:

This document captures all the features that were proposed during the project planning phase and tracks whether each feature was successfully implemented and demonstrated. It serves as evidence of the team's ability to deliver on the proposed solution.

S.No	Feature Name	Description	Status (Implemented / Partial / Pending)	Demonstrated (Yes / No)	Remarks
1	Machine Learning Inference Engine	Preprocessing input variables using scale.pkl (StandardScaler) and scoring them using rdf.pkl (XGBoost Classifier) to determine loan approval.	Implemented	Yes	Integrated successfully in /submit POST route, producing robust predictive outcomes.
2	Dynamic Loan Burden Calculator	Computes the annual loan-to-income burden percentage client-side as numerical fields are typed: $\text{Ratio} = (\text{LoanAmt} * 1000) / ((\text{AppInc} + \text{CoappInc}) * 12) * 100$.	Implemented	Yes	Triggers on inputs, rendering real-time metrics inside the burdenMeter UI block.
3	Profile Hint Engine	Analyzes the combination of credit record history, total income, and requested amount to output a pre-screening feedback tip.	Implemented	Yes	Provides alerts such as warning of high amount with weak credit history directly in the form view.
4	Interactive Scenario Presets	Injects mock candidate profiles (low-risk vs. high-risk parameters) into all 11 fields instantly on clicking preset buttons.	Implemented	Yes	Simplifies system debugging and visual presentations during evaluations.
5	Form Draft Local Storage	Enables saving the current inputs to browser memory (localStorage) and reloading drafts upon returning to the site.	Implemented	Yes	Restores the form structure using the key smartLenderDraft V1 without database calls.
6	Prediction Dashboard (Confidence & Risk Bands)	Displays approved/rejected statuses accompanied by calculation probability (%) and risk tier grouping.	Implemented	Yes	Maps XGBoost confidence metrics dynamically into Low Risk, Moderate Risk, and High Risk bands.

Feature Implementation Summary:

Total Features Proposed	6
Total Features Implemented	6
Total Features Demonstrated	6
Overall Implementation Rate (%)	100%